

draft_analysis

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Upload and Clean Data

PFR Draft Data

```
drafts = load_draft_picks(seasons = 1994:2016)

# start in 1994 since that's when NFL draft first had the current seven-round format
# end in 2015 since we want players with >= 10 seasons of experience

draft_data_raw = drafts %>%
  select(season, round, pick, pfr_player_name, position, category, side, team, w_av, dr_av) %>%
  mutate(
    team = ifelse(team == "RAI", "LVR", team),
    team = ifelse(team == "OAK", "LVR", team),
    team = ifelse(team == "RAM", "LAR", team),
    team = ifelse(team == "STL", "LAR", team),
    team = ifelse(team == "SDG", "LAC", team),
    round = factor(round, levels = 1:7),
    draft_day = case_match(
      round,
      "1" ~ "Day 1",
      "2" ~ "Day 2",
      "3" ~ "Day 2",
      "4" ~ "Day 3",
      "5" ~ "Day 3",
      "6" ~ "Day 3",
      "7" ~ "Day 3"
    ),
    category = ifelse(category == "K" | category == "P" | category == "LS", "ST", category),
    category = factor(
      category,
      levels = c("QB", "RB", "WR", "TE", "OL", "DL", "LB", "DB", "ST")),
    side = case_match(
      side,
      "O" ~ "Offense",
      "D" ~ "Defense",
      "S" ~ "Special Teams"
    ),
    side = factor(side, levels = c("Offense", "Defense", "Special Teams")),
    w_av = ifelse(is.na(w_av), 0, w_av) # NA means no NFL snaps; therefore 0 AV,
  )
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `draft_day = case_match(...)`.
```

Caused by warning:

```
## ! `case_match()` was deprecated in dplyr 1.2.0.
## i Please use `recode_values()` instead.
```

```
write_csv(draft_data_raw, "Data/draft_data_consolidated_1994_2016.csv")
```

Draft Value Charts

```
draft_value_meers = read_excel("Data/nfl_draft_value_charts.xlsx", sheet = "DVC_New_Raw") %>%
  mutate(
    value = format(value, nsmall = 1),
    value = round(as.numeric(value), 1)
  ) %>%
  rename(meers_value = value)

draft_value_johnson = read_excel("Data/nfl_draft_value_charts.xlsx", sheet = "DVC_Original_Raw") %>%
  mutate(
    value = format(value, nsmall = 1),
    value = round(as.numeric(value), 1)
  ) %>%
  rename(johnson_value = value)
```

Merged Data

```
draft_data = draft_data_raw %>%
  left_join(draft_value_meers, by = "pick") %>%
  left_join(draft_value_johnson, by = "pick") %>%
  mutate(
    efficiency_overall = round(w_av / meers_value, 3),
    efficiency_team = round(dr_av / meers_value, 3),
    efficiency_overall_johnson = round(w_av / johnson_value, 3),
    efficiency_team_johnson = round(dr_av / johnson_value, 3),
  )
```

Exploratory Data Analysis

Total Weighted AV by Draft Season

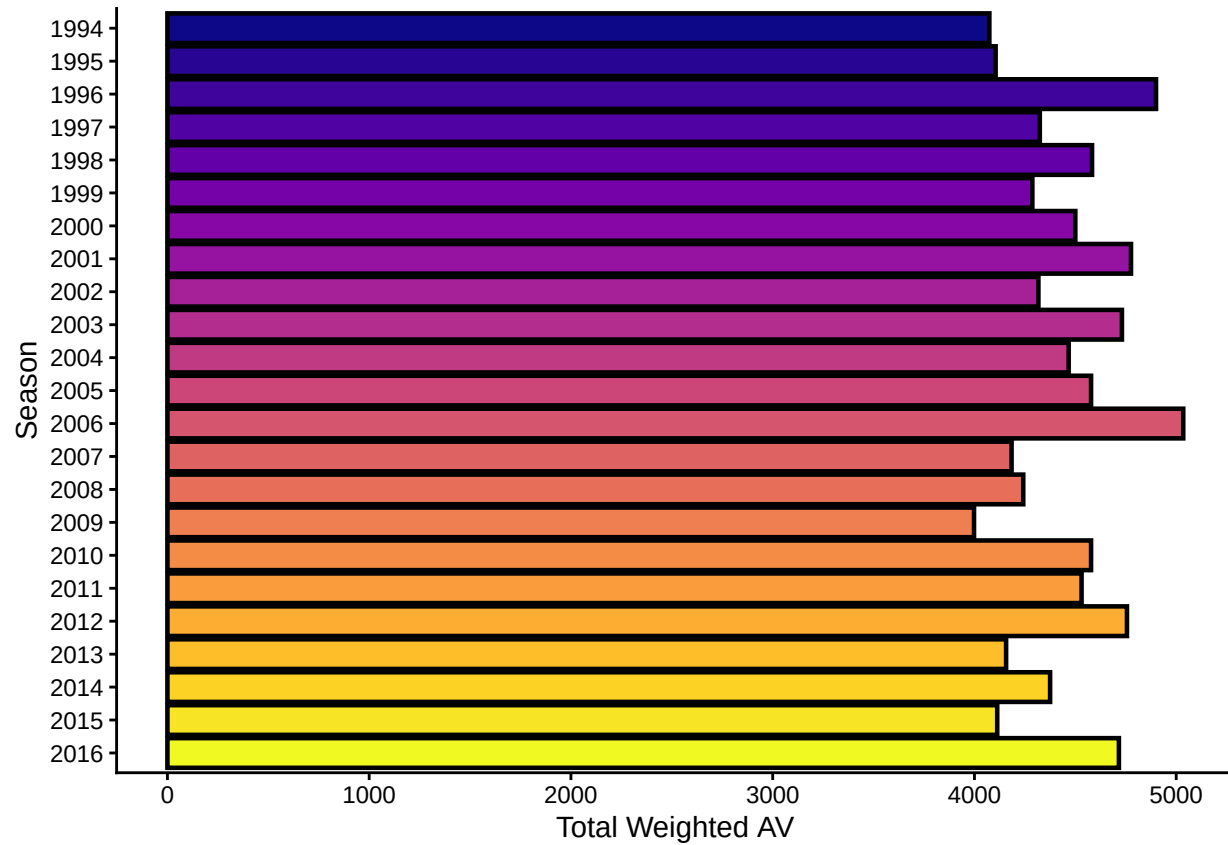
```
basic_av_plot = draft_data %>%
  group_by(season) %>%
  summarize(total_av = sum(w_av, na.rm = TRUE)) %>%
  mutate(season = factor(season, levels = rev(1994:2016))) %>%
  ggplot(aes(x = season, y = total_av, fill = season)) +
  geom_bar(stat = "identity") +
  geom_col(color = "black", linewidth = 0.8) +
  coord_flip() +
  scale_fill_viridis_d(option = "C", direction = -1) +
  theme_classic() +
  theme(legend.position = "none") +
  labs(
```

```

y = "Total Weighted AV",
x = "Season"
)

```

basic_av_plot

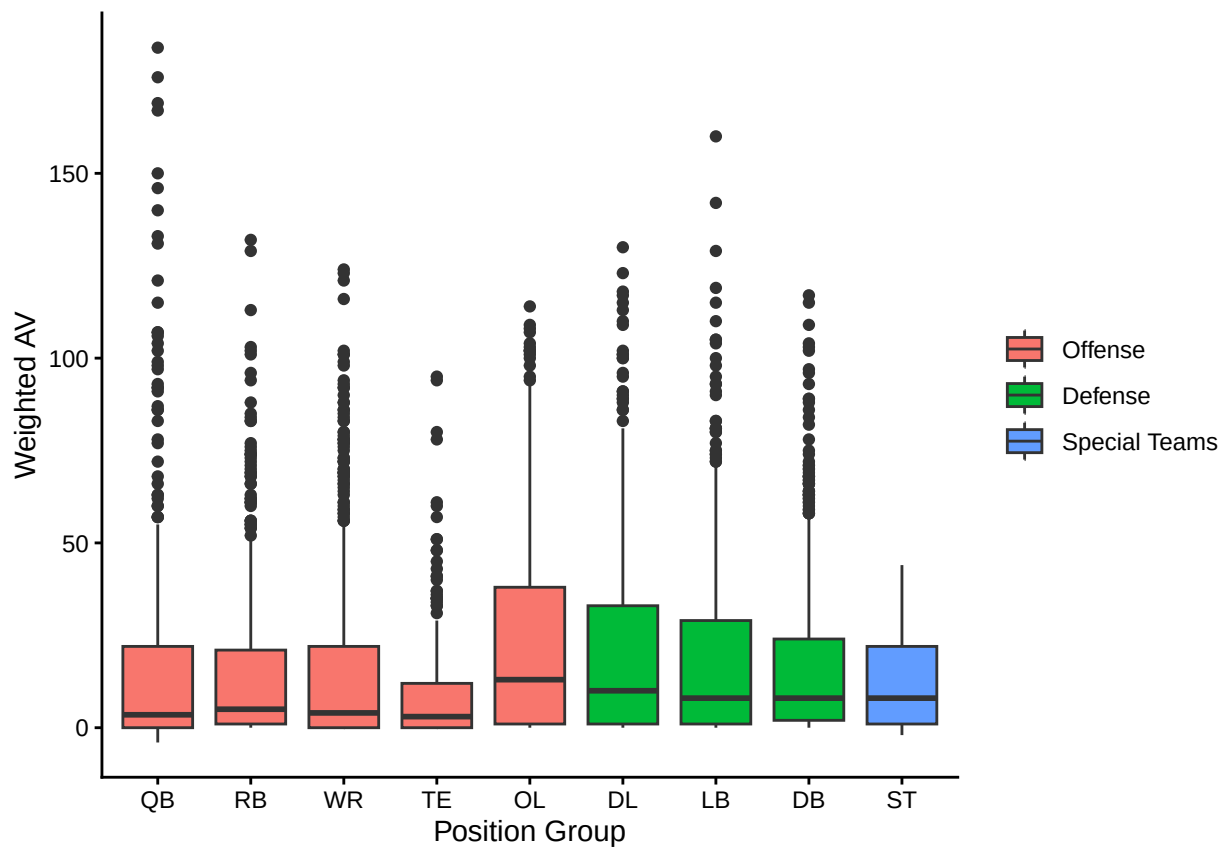


Weighted AV Per Player, by Position Category

```

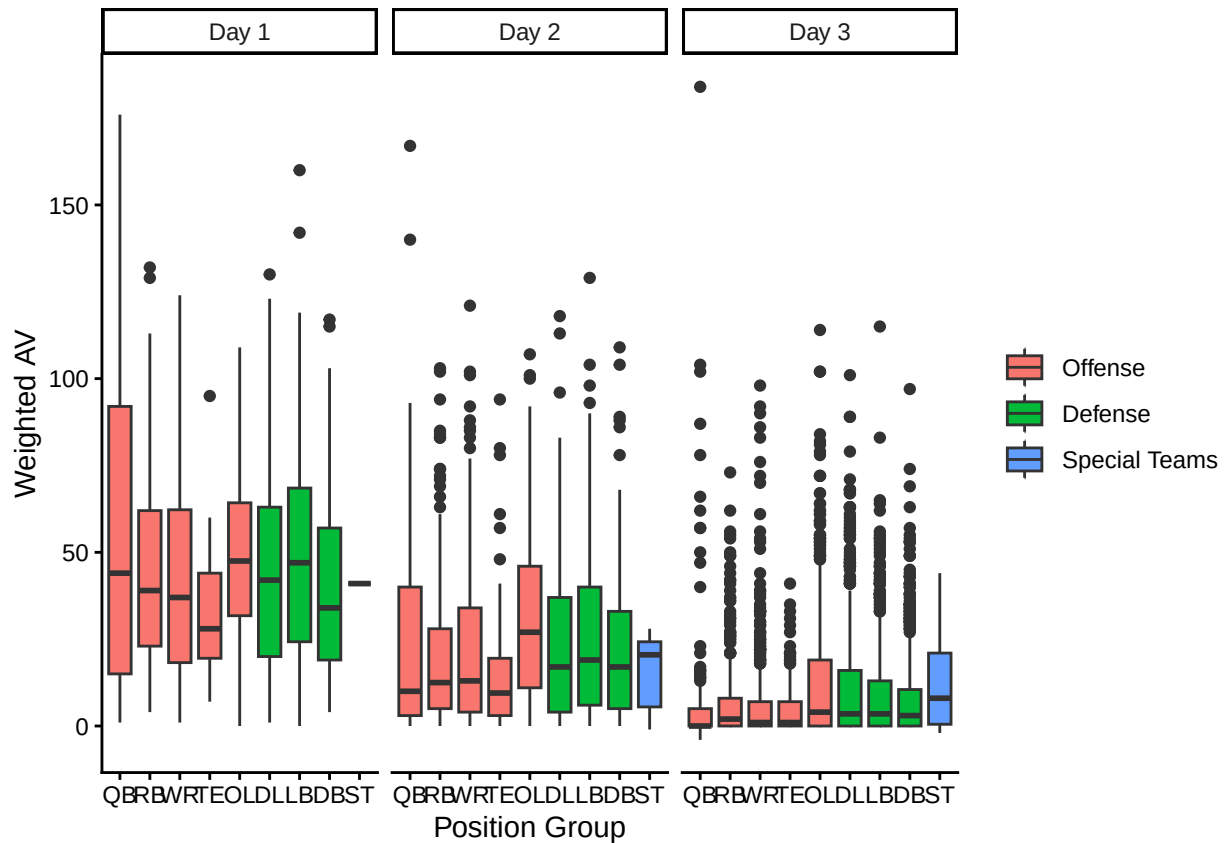
draft_data %>%
  ggplot(aes(x = category, y = w_av, fill = side)) +
  geom_boxplot() +
  theme_classic() +
  labs(
    y = "Weighted AV",
    x = "Position Group"
  ) +
  theme(legend.title = element_blank())

```



Weighted AV Per Player, by Position Category, by Day

```
draft_data %>%
  ggplot(aes(x = category, y = w_av, fill = side)) +
  geom_boxplot() +
  theme_classic() +
  labs(
    y = "Weighted AV",
    x = "Position Group"
  ) +
  facet_wrap(~ draft_day) +
  theme(legend.title = element_blank())
```



Scatterplots of Draft Value vs. Weighted AV

```
scatter_johnson = draft_data %>%
  group_by(pick) %>%
  summarize(mean_av = mean(w_av, na.rm = TRUE)) %>%
  ungroup() %>%
  left_join(draft_value_meers, by = "pick") %>%
  left_join(draft_value_johnson, by = "pick") %>%
  ggplot(aes(x = johnson_value, y = mean_av)) +
  geom_point() +
  geom_smooth(method = "lm", col = "firebrick") +
  geom_smooth() +
  theme_classic() +
  labs(
    title = "Johnson Method",
    x = "Pick Value (Johnson's Method)",
    y = "Mean Weighted AV"
  )

scatter_meers = draft_data %>%
  group_by(pick) %>%
  summarize(mean_av = mean(w_av, na.rm = TRUE)) %>%
  ungroup() %>%
  left_join(draft_value_meers, by = "pick") %>%
  left_join(draft_value_johnson, by = "pick") %>%

```

```

ggplot(aes(x = meers_value, y = mean_av)) +
  geom_point() +
  geom_smooth(method = "lm", col = "firebrick") +
  geom_smooth() +
  theme_classic() +
  labs(
    title = "Meers Method",
    x = "Pick Value (Meers' Method)",
    y = "Mean Weighted AV"
  )
)

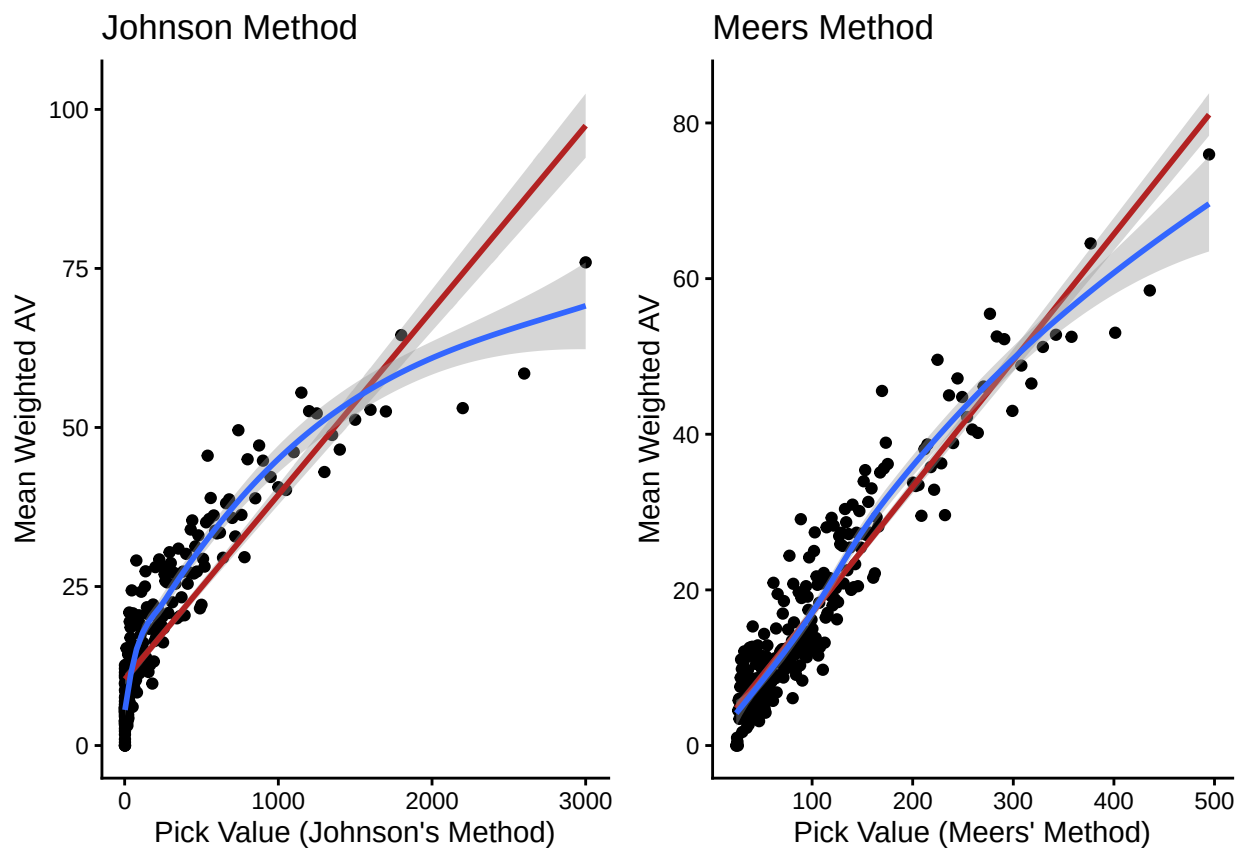
plot_grid(plotlist = list(scatter_johnson, scatter_meers))

```

```

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'

```



Scatterplots of Draft Value vs. Weighted AV, By Position Group

```

scatter_johnson_pos = draft_data %>%
  group_by(pick, category) %>%
  summarize(mean_av = mean(w_av, na.rm = TRUE)) %>%
  ungroup() %>%
  left_join(draft_value_meers, by = "pick") %>%

```

```

left_join(draft_value_johnson, by = "pick") %>%
ggplot(aes(x = johnson_value, y = mean_av)) +
geom_point() +
geom_smooth(method = "lm", col = "firebrick") +
geom_smooth() +
theme_classic() +
facet_wrap(~ category) +
labs(
  title = "Johnson Method",
  x = "Pick Value",
  y = "Mean Weighted AV"
)

```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by pick and category.
## i Output is grouped by pick.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(pick, category))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

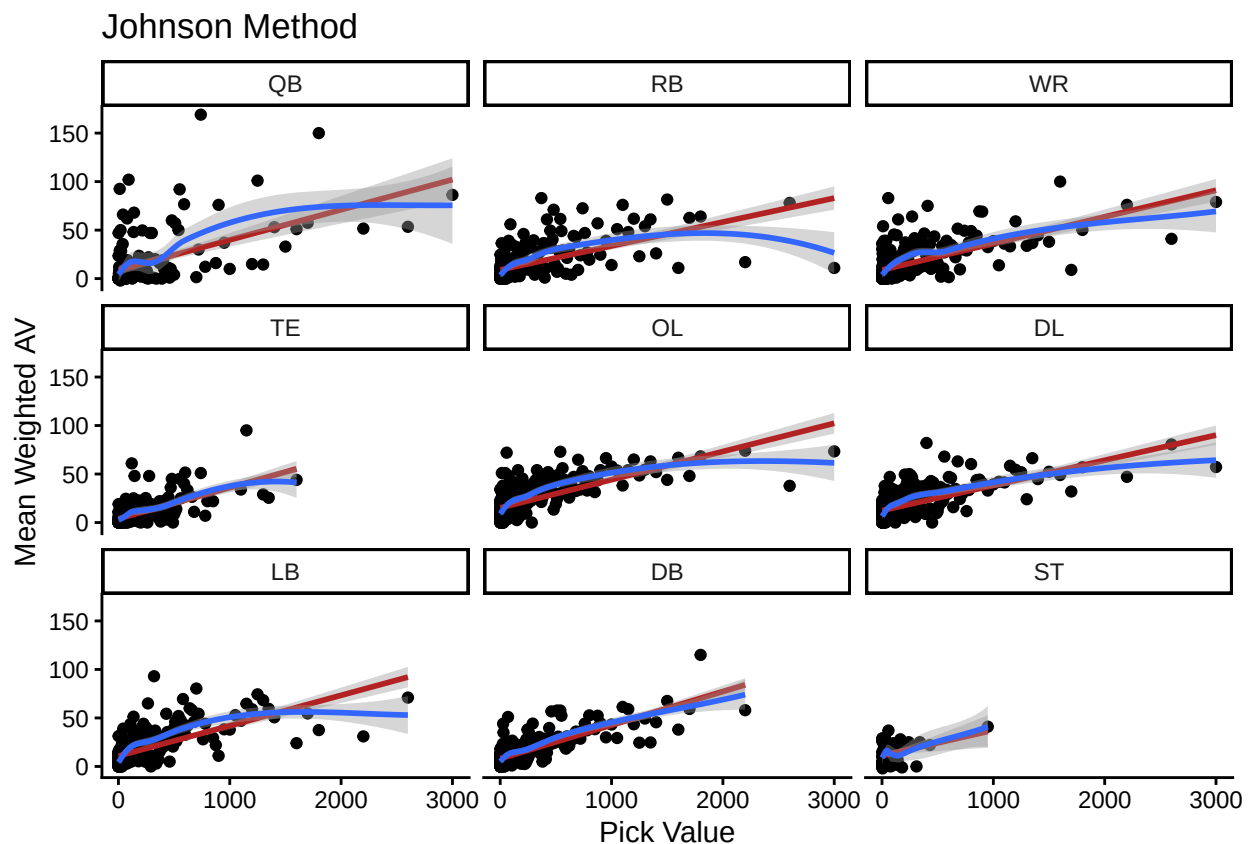
```

```
scatter_johnson_pos
```

```

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'

```



```

scatter_meers_pos = draft_data %>%
  group_by(pick, category) %>%
  summarize(mean_av = mean(w_av, na.rm = TRUE)) %>%

```

```

ungroup() %>%
left_join(draft_value_meers, by = "pick") %>%
left_join(draft_value_johnson, by = "pick") %>%
ggplot(aes(x = meers_value, y = mean_av)) +
geom_point() +
geom_smooth(method = "lm", col = "firebrick") +
geom_smooth() +
theme_classic() +
facet_wrap(~ category) +
labs(
  title = "Meers Method",
  x = "Pick Value",
  y = "Mean Weighted AV"
)

```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by pick and category.
## i Output is grouped by pick.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(by = c(pick, category))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

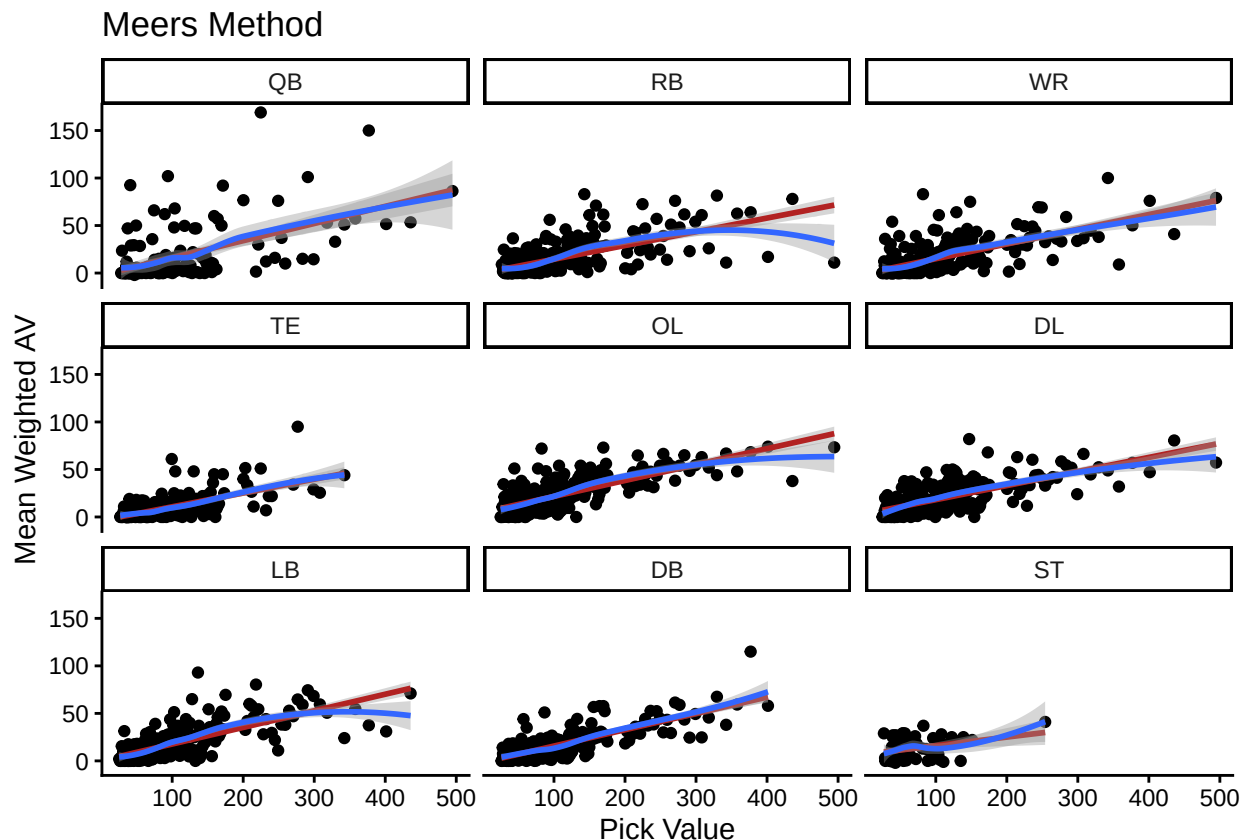
```

```
scatter_meers_pos
```

```

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'

```



Building a Shiny Dashboard!

```
draft_clean <- draft_data %>%
  transmute(
    season,
    team,
    round,
    pick_overall = pick,
    player = pfr_player_name,
    position,
    category,
    side,
    draft_day,

    # AV inputs
    w_av,
    dr_av,

    # draft value models
    meers_value,
    johnson_value,

    # precomputed draft value efficiencies
    efficiency_overall, # w_av / meers_value
    efficiency_team,    # dr_av / meers_value
    efficiency_overall_johnson,
    efficiency_team_johnson
  ) %>%
  mutate(category = as.character(category))

write_rds(draft_clean, "Data/draft_shiny.rds")

team_colors <- c(
  "ARI" = "#97233F",
  "ATL" = "#A71930",
  "BAL" = "#241773",
  "BUF" = "#00338D",
  "CAR" = "#0085CA",
  "CHI" = "#0B162A",
  "CIN" = "#FB4F14",
  "CLE" = "#311D00",
  "DAL" = "#003594",
  "DEN" = "#FB4F14",
  "DET" = "#0076B6",
  "GNB" = "#203731",
  "HOU" = "#03202F",
  "IND" = "#002C5F",
  "JAX" = "#006778",
  "KAN" = "#E31837",
  "LVR" = "#000000",
  "LAC" = "#002A5E",
  "LAR" = "#0B0B0B",
  "MIA" = "#008E97",
```

```
"MIN" = "#4F2683",  
"NWE" = "#002244",  
"NOR" = "#D3BC8D",  
"NYG" = "#0B2265",  
"NYJ" = "#125740",  
"PHI" = "#004C54",  
"PIT" = "#FFB612",  
"SEA" = "#002244",  
"SFO" = "#AA0000",  
"TAM" = "#D50A0A",  
"TEN" = "#4B92DB",  
"WAS" = "#5A1414"  
)
```